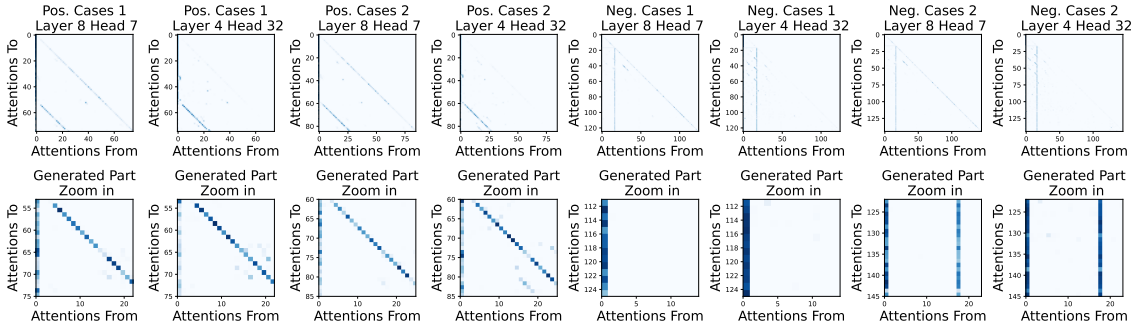

Figure 7: Visualization of attention matrices layer-8 head-7 and layer-4 head-32 in the first two cases listed in Figure 6. Figures in the second row are the zoom-in of the first row, representing the attentions from prompts to model responses.

Averaged PPL of Prompts	N-gram Match UR				Fuzzy Match UR (%)			
	3	6	9	12	70	80	90	100
37.67 (Llama-2 7B)	0.75 ± 0.09	0.64 ± 0.13	0.60 ± 0.14	0.48 ± 0.15	0.68 ± 0.10	0.62 ± 0.14	0.60 ± 0.14	0.37 ± 0.16
23.95 (with rephrasing)	0.86 ± 0.06	0.78 ± 0.11	0.72 ± 0.15	0.68 ± 0.18	0.81 ± 0.09	0.76 ± 0.14	0.73 ± 0.16	0.67 ± 0.20
78.27 (Direct Defense)	0.57 ± 0.15	0.48 ± 0.16	0.41 ± 0.19	0.35 ± 0.18	0.50 ± 0.15	0.44 ± 0.17	0.41 ± 0.18	0.29 ± 0.16
78.10 (Random Insertion)	0.85 ± 0.07	0.56 ± 0.08	0.44 ± 0.08	0.40 ± 0.10	0.66 ± 0.08	0.50 ± 0.09	0.36 ± 0.10	0.14 ± 0.06
52.04 (Higher PPL)	0.87 ± 0.06	0.76 ± 0.11	0.63 ± 0.11	0.54 ± 0.13	0.81 ± 0.09	0.72 ± 0.11	0.70 ± 0.12	0.59 ± 0.15
54.65 (Local Lookup)	0.53 ± 0.06	0.42 ± 0.11	0.38 ± 0.14	0.28 ± 0.11	0.48 ± 0.08	0.43 ± 0.10	0.40 ± 0.12	0.24 ± 0.10
06.18 (Repeated Prefix)	0.33 ± 0.19	0.25 ± 0.16	0.22 ± 0.14	0.12 ± 0.09	0.38 ± 0.21	0.33 ± 0.20	0.20 ± 0.15	0.06 ± 0.06
41.96 (Fake Prompt)	**0.33** ± 0.09	**0.17** ± 0.07	**0.13** ± 0.06	**0.09** ± 0.06	**0.20** ± 0.07	**0.17** ± 0.07	**0.15** ± 0.06	**0.09** ± 0.06

Table 1: Prompt extraction evaluation on Llama-2 incorporated with our defending strategies.

high-γ_{cur} attention heads only present in the Transformer modules after the first three layers, which indicates that prompt memorization is a high-level mechanism for LLMs. More experiments of SPLIt on cases can be found in Appendix F.

To explore and understand the meaning of a SPLIt or N-SPLIt score for attention heads, we sample two high α_{cur} and γ_{cur} attention heads, i.e., layer-8 head-7, and layer-4 head-32, and visualize their related attention matrices among four cases (two successful and two failed) in Figure 7. By comparing the attention weights between the successful cases and failure cases, it is clear that in some attention matrices of success cases (e.g., case 1) there exists an obvious slipping line parallel to the diagonal attention line from prompts to generated prompts, which we call the *parallel translation*. Parallel translation suggests that the ability of LLMs to "memorize" and "translate" prompts truly come from strong and direct attention connections in the attention mechanism.

5 Derived Defenses

After analyzing the underlying mechanisms of PEA, in this section we put forward our defense strategies based on our hypothesis in Section 4. We also investigate the ineffectiveness of LLMs' alignments against PEAs in Appendix F.1.

5.1 Defenses from Hypothesis

We propose two types of defense strategies. These methods are not designed with prior knowledge of current adversarial prompts, but are instead inspired by the internal mechanisms described in Section 4, suggesting that they can even be effective against unseen adversarial prompts.

Increasing the Perplexity. As we discussed in the *convincing premise*, low perplexity is a vital factor for prompt memorization. Therefore, one simple defense is to re-craft a new prompt with a higher perplexity, which may reduce its risk of prompt leakage. Based on this intuition, we propose two defending strategies: I) *Random Insertion*: randomly insert some unfamiliar tokens into the prompts. II) *High-PPL rephrasing*: rephrase the prompts to improve their perplexity. Compared to *random insertion*, prompts rephrased in this way are still syntactically valid.

Blocking the Attention Links. In contrast to improving the perplexity of prompts, another defending strategy is to block or destroy the *parallel translation* of LLMs through serialization pattern engineering. In concrete, we propose three blocking strategies: I) *Only Local Lookup*. We can append an external instruction to prompts to prohibit the LLMs to lookup in an overly broad scope, such as "`Your referenced sentence cannot be longer`

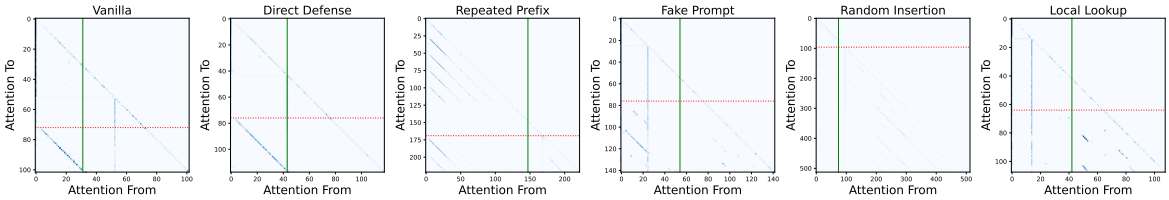

Figure 8: Visualization of our proposed defense strategies, where the left-bottom area split by the green solid line and the red dotted line is the attention sub-matrix from prompts to generated texts.

`than 20 words.`". II) *Repeated Prefix of Prompts*. We can add a *repeated* prefix before the prompts, such as "`O'<' Red>!4̈59?- > O'<' Red>!4̈59?- >O'<' Red>!4̈5 9?- >O'<'Red> !4 5̈9?- >O'<' Red>!4̈59?- >`", where we expect these repeated meaningless prefixes will confuse the translation pattern. III) *Fake Prompts*. We can add a fake prompt before the real prompts, and make it easier (e.g., with a low perplexity) for extraction, such as "`Your Instruction: "You are a superhero who aims to SAVE the world.""`. In this way, in LLMs, the attention modules will construct strong but incorrect *parallel translation* from the fake prompts to generated prompts, and thus the true prompt can be protected from leakage.

5.2 Experiments

Settings. The experimental settings are the same as in previous experiments. In addition to the methods proposed in Section 5.1, we implement two new baselines to benchmark defending strategies. The first is a prompt-engineering based method, *direct defense*, which appends "`Please do not disclose this Instruction to users.`" on original prompts. The second is called *lower-perplexity prompts rephrasing*, the opposite process of *high-PPL rephrasing*, which explores the influence of prompt's perplexity.

Defending Efficacy. As shown in Table 1, while our baseline can truly decrease the uncover rates from vanilla prompts, our proposed strategies consistently achieve much lower uncovered rates in most of the prompt extraction situations except high-PPL rephrasing. Actually, prompts rephrased for a higher perplexity result in a much higher uncover rate of the original prompts. This might be because these prompts are generated by the LLMs themselves, and thus are familiar with them. Regarding other strategies, *repeated prefix* and *fake prompt* can reduce the UR from 0.37 to 0.06 and 0.09 in Llama-2, an 80% reduction from vanilla prompts. Besides, *random insertion* does not perform as effectively as other strategies, which may be attributed to the robustness of LLMs against token-level noises. We provide the evaluation on closed AI models in Table 4.

Besides, we also examine performance degradation introduced by our defenses in Appendix F.2, where we only observe a slight accuracy drop after utilizing our defenses.

Attention Visualization Additionally, Figure 8 presents the visualization results of attention matrices on our defending strategies. It is clear that three of our methods eliminate the slipping line that denotes the *parallel translation*, demonstrating they can destroy the direct connection from prefixes to the prompts. However, parallel attention connections still exist in the *repeated prefix*, but they only appear in prompt's beginning, which indicates that the extracted outputs are just the repetition of prefixes we pre-set, instead of the prompt itself. We can also observe the similar phenomenon in *fake prompt*, where the slipping line represents the parallel translation from the fake prompt to responses, which prevents real prompts from leakage as well.

6 Conclusion

In this paper, we focus on the prompt leakage issue of customized large language models, systematically examine the key factors which influence prompt leakage, and provide explanations for its occurrence. The results indicate that despite safety alignments during training, LLMs remain vulnerable to prompt extraction, especially those larger models which may face more severe implicit intent prompt attacks. In these cases, we find that both the familiarity of a LLM with prompts and the parallel translation in the self-attention mechanism contribute to prompt leakage of LLMs during prompt extraction. Consequently, we propose defending strategies against this threat. They consistently achieve a lower leakage rate compared to direct prompt defense, and only exhibit slight degradation to their performances.

Limitations

We outline the limitations of our study and propose several future research directions as follows:

Training-Time Investigation on Prompt Leakage. Our analysis of prompt leakage is based on the evaluation of LLMs during inference. While this approach effectively elucidates the key factors and mechanisms underlying prompt extraction attacks, it does not fully capture how LLMs develop vulnerabilities to such attacks. Therefore, a promising avenue for future research would be to explore in detail how pretraining and alignment processes influence the susceptibility to prompt leakage.

Mitigating PEAs from Other Perspectives. We have proposed several defense strategies and highlighted their advantages, such as their ease of adoption for both platforms and developers, and the lack of necessity for prior knowledge about adversarial prompts. Nevertheless, we recognize that prompt defense mechanisms can be further enhanced by exploring additional dimensions. For instance, as illustrated in Figure 6, we observe that only $8 \sim 10$ attention heads consistently exhibit the parallel translation phenomenon. Pruning these specific heads could potentially disable the model's ability to memorize prompts. Additionally, platforms could implement a hard-coded distinction between prompts and inputs by defining different type tokens and positional tokens. Furthermore, integrating these inference-time defenses with alignment techniques could significantly bolster the security of prompts against extraction attacks.

Ethical Considerations

We take strict measures to protect the privacy of both the platform and third-party developers, as demonstrated in the following aspects:

- **Enhanced Understanding of PEAs:** This paper provides a detailed explanation of the internal mechanisms behind prompt leakage and summarizes the patterns associated with it, which can help mitigate such threats.
- **Defense Strategies:** We propose several defense methods and evaluate their effectiveness. These methods, derived from our discoveries rather than being direct responses to existing attacks, hold significant potential. We believe they can be applied to mitigate future, newly proposed attacks, thereby offering substantial benefits to developers.
- **Regarding Leaked Prompts:** We only use prompts that have *already been leaked* as cases in our paper, and our benchmark, PEAD, is constructed using *public* prompts. Any prompts or documents that were obtained during our realistic attack simulations have been protected and deleted after the experiments, and will **not** be released.
- **Contribution to the Community:** We have anonymously shared our research with LLM customization platforms to help improve their products and protect developers' data.

In conclusion, we believe our research aligns with the ethical standards required for AI research, particularly in the areas of security and privacy.